

Lawbot: An Enhanced Legal Information Retrieval System using RAG

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Abstract—A large section of the public still remains legally illiterate. This lack of legal education is more common among low-income groups, marginalized communities, and the rural population. Furthermore, the absence of understandable legal information delays the judicial process which adds to the backlog of millions of pending cases. This paper presents Lawbot, a Retrieval-Augmented Generation (RAG)-based legal information retrieval system designed to improve public access to legal knowledge. Integrating the Kanoon API for authentic case retrieval and embedding-based semantic search, Lawbot provides modules for chat-based queries, document summarization, and real-time legal news and blogs. The system achieved high accuracy for user satisfaction in survey-based evaluations. Compared to existing Lawbot demonstrates superior contextual understanding and explainability through RAG-driven augmentation. The results highlight its potential as an accessible and scalable solution for enhancing legal literacy and democratizing legal information access.

Keywords—Document Summarization, Information Retrieval, Legal Literacy, Natural Language Processing, RAG.

I. INTRODUCTION

The Indian legal framework, one of the most sophisticated in the globe, caters to a population of more than 1.4 billion citizens. Yet, a large number of the population are legally illiterate, without knowledge of their rights, legal remedies, and due processes. Such ignorance and lack of knowledges brings down people of rural masses, women, and marginalized classes as they are often taken advantage of. They are often used by moneylenders, bribe-seeking officials, and intermediaries. Lacking sufficient legal consciousness, people cannot exercise their rights, protect themselves against discriminatory legal actions, or enter the legal system with efficiency. Legal literacy in India is still remedial, not preventative, as citizens come to know their rights only after a legal violation has occurred. Estimates indicate that some 70% of rural Indians are illiterate, with even more having no knowledge of their legal rights and protections. [1].

Over 80,000 cases are pending in the Supreme Court, and millions more in the High Courts and District Courts, many that stem from preventable legal misunderstandings [2]. It is seen that most legal conflicts can be resolved by mediation or early legal action instead of becoming protracted to court fights because of ignorance. The absence of legal education at the school level also contributes to this crisis, as most people only experience the legal system after they are involved in conflicts.

To address this issue, interactive legal literacy programs have emerged as a potential solution. Paper [3], discusses the legal illiteracy in India. It provides interactive legal literacy sessions as a solution to this problem. These sessions, Panchayat Shivir, focused on tribal communities in Rajasthan, where people struggle to understand the Tribal Self-Rule Law, which governs local governance and resource management. These initiatives interact directly with the people in order to close the gap between themselves and the intricacies of the legal world. This leads to a more legally informed population.

As stated by [4], normal citizens of the nation who are educated but not lawyers believe legal terminology is alien to them since it is too difficult to read or interpret without professional assistance.

Legal literacy is an important problem in this country because most people are denied access to reliable, understandable, and location specific information on the laws of the country. Common legal materials, including official government websites, law textbooks, and face-to-face meetings with lawyers tend to show a necessity of possessing a high degree of legal education, economic means, or access to legal experts. This would be a major impediment for people who require legal assistance but do not possess the ability to obtain it. Our suggested AI-driven chatbot fills this void by providing affordable, real-time legal support in the areas of consumer rights, property and finance conflicts.

II. LITERATURE SURVEY

A. Comparative Analysis of Legal Chatbots

Various efforts have been made to enhance legal literacy and access among unaware legal proceedings in India, ranging from community-level efforts to AI-based assistive solutions. Work done in the field is discussed in this section, ranging from interactive legal awareness programs to AI-based legal assistants, along with their merits and demerits.

1) *NyayGuru*: NyayGuru [5], an AI-powered legal assistant, enhances legal literacy and access in India by providing multilingual and legal advice at a fairly low price. It offers 24/7 assistance being a chatbot, fosters preventive legal literacy and helps in reducing dependency on legal intermediaries or middle-men. But it is handicapped in terms of the absence of case-sensitive advice, its possible unreliability, disadvantage of digital inaccessibility, and

ethical dilemma. It can provide generic guidance without knowing definite contexts.

2) *IndiaGPT*: Supported by context-based AI, IndiaGPT [6] guides the users towards legal rights, procedures, and case laws. But it is limited in its application. It lacks any sort of tailored or case-specific advice, and relies heavily on AI provided responses. It is also vulnerable to online accessibility issues, hence becoming a secondary tool and not a replacement for legal professionals. It also uses old information making its advice less valuable.

3) *LawbotPro*: LawbotPro [7] is an AI-based legal software with experience in automated legal document preparation, case law research, and procedural guidance. It helps the users to write legal contracts, research precedents, and be aware of the procedures of courts, thus facilitating easier handling of law. LawbotPro promotes access to justice via low-cost, artificial intelligence-driven legal services, reducing dependency on expensive legal services. It is vulnerable for including potential AI bias, lack of human judgment, and it faces a challenge in the determination of complex cases of law. Its demerits surely validate the need for human control over legal matters. LawbotPro does not utilize exact legal sources or historical case precedents.

We can see how different AI-powered chatbots have emerged in the legal sphere, all of which attempt to address legal literacy by robot-legal advice. However, these alternatives have varying limitations on how far they can provide legally valid, jurisdictionally appropriate, and user-appropriate legal advice. Table IV includes a comparison of existing legal chatbots and the system proposed here.

B. Related Work

There are numerous researches undertaken to process case data in order to make judgement prediction depending on various suggested framework. An Indian Virtual Legal Assistant [8], is an AI-based system that will help users in legal questions with the help of analyzing legal cases in the past and give suggestions. The VLA is divided into four primary components: Text Analytics, Knowledge Base, Question Generation Engine and Chabot Interface with Speech Recognition. They help it understand, process, and answer legal questions appropriately. It ranks cases in a decreasing order based on relevance and employs a KNN-based method to categorize legal cases.

A Chinese legal judgement prediction [9] in which researchers incorporate physiological attributes and fact description of incidents to forecast the judgement. The Chinese Legal Consultation System (CLCS), [10], combines legal ontology and artificial intelligence (AI) to improve legal information retrieval. The system builds a legal knowledge base upon both statutes ontology (top-down approach) and case ontology (bottom-up approach), enabling for semantic understanding and legal case retrieval.

Paper [11] suggests an ontology-based knowledge block summation approach for Chinese judgment classification. It identifies principal legal concepts with the help of ontologies and improves classification with Word Mover's Distance (WMD) and K-Nearest Neighbors (KNN) for better efficiency and accuracy.

Another work [14] constructs a use case of GNNs legal cases as nodes, connected with citations, laws, and time for

classification and link prediction. GraphSAGE is utilized for binary classification of case outcomes and relation prediction among cases. Various text embeddings (random, XLNet, hierarchical) and edge types (directed, reverse-directed, undirected) were tested for better performance.

Another Chatbot-based IR [13] employs FAQ datasets and StarSpace for intent classification. Response accuracy is enhanced through supervised learning using StarSpace and word embeddings. BERT fine-tunes a multilingual legal Q&A model, boosted by a hybrid IR + NLP process. Although [12] learned its chatbot from legal documents, with legal expert-created Q&A pairs and ChatGPT, LlamaIndex served to perform fast indexing of large legal texts and vectorization for retrieval. Assessed with 50 legal queries, having performance measures comprising accuracy (82%) and F1-score (79%).

TaxGuru [15] critiques the long-standing inaction in creating a High Court Bench in Western Uttar Pradesh, shedding light on how more than 10 crore citizens have significant geographical obstacles (700-800 km distance) to reach justice. It denounces the central government's indifference despite seven decades of clamor, focusing on systemic disparities in India's judicial infrastructure. Ghosh et al. [16] discusses conflicts between India's proposed Uniform Civil Code and tribal customary laws in the Northeast, arguing that blanket uniformity risks eroding tribal autonomy. This article suggests that the best approach for developing legal standards is to create systems that are mindful of both constitutional requirements and local variations.

Smith et al. [17] performed a benchmarking analysis of LLMs on how well they perform in regard to making correct legal analyses in the sense of being able to determine how to apply the law to new sets of facts to those sets of facts. Paper [18] looks at the lack of unified regulatory frameworks for the use of AI in the fintech sector in India, where identified deficiencies in the Digital Personal Data Protection Act (2023) are outlined and proposed establishment of sector-specific compliance protocols aimed at reconciling innovation and consumer protections for AI-facilitated solutions. In Paper [19], the authors describe the importance of transparency when developing legal AI systems, and based on this importance, recommend the establishment of processes to perform bias audits and to develop legal AI systems with transparency built in. Paper [20] explores a study regarding the efficiency gains produced by using AI-guided profiling of clients in legal technology. Paper [21] explores the potential compliance risks of AI personalization as they relate to the implementation of the General Data Protection Regulation and the California Consumer Privacy Act. [21] also points out the legal conflicts and jurisdictional issues created by processing personal data outside the borders of any country, and as a result, recommends that data from outside the confines of a single jurisdiction be encrypted and standardized in their reporting format to minimize any potential legal risk.

ACC has released a report stating that AI domain-specific tools resulted in 60% less spending on legal research. It reported that domain-specific modules performed better than generic models; however, all models require routine monitoring by a human [22]. Wang et al. [23] created a benchmark called "LegalBench" to measure the legal reasoning ability of 20 different language models on 162 different tasks. It found that the trend for these models was to have a greater recall of statutory law; however, they all scored lower than other models when it came to ethical judgments. A

publication referenced in [24] contrasted the extraction vs. abstracting of legal documents and determined that the best approach to produce accurate summaries was a combination of the two.

Zakir et al. [25] provided a comprehensive overview of artificial intelligence (AI) in legal research; the study determined that by using AI, practitioners could retrieve case law 50% faster than previously possible and warned of ethical issues surrounding the technology, such as the surveillance and bias in training data. Similarly, research referenced in [26] advocated for implementing pro bono clinics as a part of law schools' education programs. The study noted that students who participated in this program improved their ability to service the public by 30% and also increased their outreach to marginalized populations. It discussed gaps in interpretability related to transformer models created for non-English text, which could create confusion for all users of these models [27]. A report referenced in [28] provided an overview of technological advancements that have been made in the field

of NLP for use in legal scholarship and therefore the future of NLP will likely involve multimodal forms of AI. Research in [29] assesses Kenya's legal empowerment initiatives, registering a 25% increase in land-rights wins for marginalized communities. Identifies digital literacy constraints hinder AI tool uptake. Research [30] examines NLP in law, with 90% accuracy in extracting contract clauses.

III. OUR APPROACH

For a solution to combat legal illiteracy and capacitate marginalized people, our strategy integrates technology solution with people-centered learning. Through creating a Legal Bot that can aid consumers and research interns in knowing their rights, the strategy is geared towards being convenient and interactive. This strategy makes use of easily accessible digital technologies and localized scenario-based learning so that legal empowerment can be adequately achieved. The subsequent sections present the specific strategies and steps that were used in this effort.

TABLE I. COMPARATIVE ANALYSIS OF LEGAL CHATBOTS

Feature	NyayGuru	IndiaGPT	LawbotPro	Proposed System
Context Awareness	Limited	Outdated Data	Moderate	High
Legal Reference Inclusion	Yes	No	No	Yes
Regional Adaptation	No	Partial	No	Yes
User Interaction	NLP-based	NLP-based	Conversational	Personalized Assistance
Case-Specific Advice	No	No	Partial	Yes

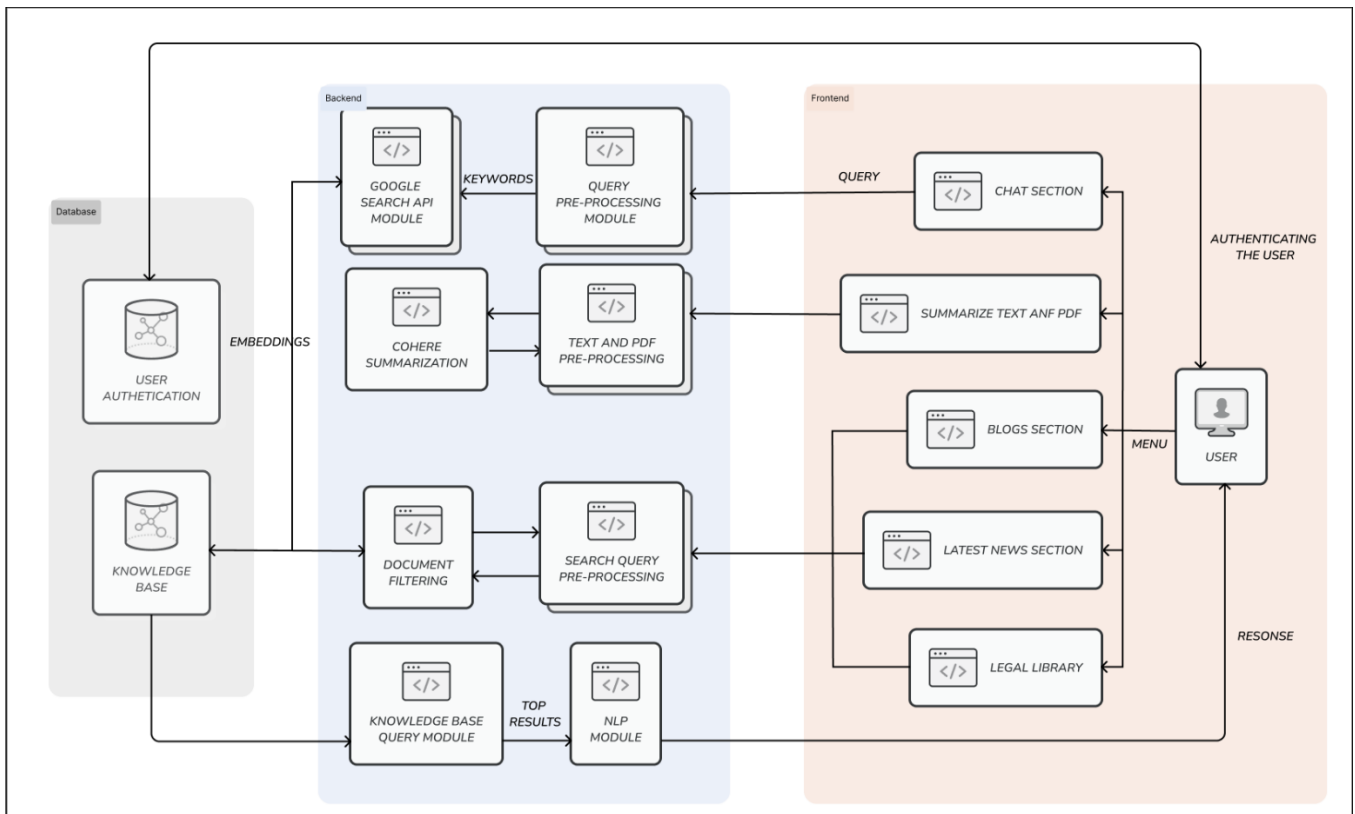


Fig. 1. System Architecture of Lawbot

Our proposed solution follows the System Architecture as shown in in Fig. 1. The architecture is built to incorporate several components that collaborate to provide context-sensitive legal advice. The primary modules are as follows:

A. User Input Processing:

The user chooses an option from the menu like chat-based queries, document summarization, blogs, news, or the legal library. The user input is preprocessed to normalize the text format and eliminate stopwords using spaCy and NLTK

tokenization methods. The input is then routed to the corresponding module depending on its intent.

B. Query Processing and Information Retrieval (RAG):

The extracted text is then passed through an LLM-based NLP model to produce well-formatted responses. BERT-based legal question-answering models are used to fine-tune responses according to jurisdictional demands. User authentication and access control are in place to ensure secure provision of personalized legal advice. The text input in question will next be sent to the Kanoon API so that legal information can be obtained externally. During this process of gathering all of the data from the dataset, each document will generate a representation of its text using a 768 dimensional vector model upon being saved in a database called FAISS. The RAG will then be used to combine the results from both the generative and extractive NLP processes with the legal information stored, producing outputs that are contextually and accurately relevant to the user making the query.

C. Legal Document Summarization:

Both Lawyers and other users routinely deal with lengthy pieces of legal documents, including Contracts, Policies and Case Law. As such, to help users read these lengthy documents quickly and gain a more comprehensive understanding of the contents, Chatbot provides automated document summarisation of the legal documents; whether they are text-based PDF's or scanned Images. PDFs are parsed using PyMuPDF to parse the document so that structure data is available to create a summary using Cohere's Abstractive Summarization Model, allowing for the generation of a Human-Like Summary. OCR-Based PDFs go under text extraction by Tesseract OCR. This is normalized, tokenized and then summarized by using extractive approaches like TextRank, BM25 for key phrase selection. It is then condensed to human understandable. This module greatly minimizes effort needed to review voluminous contracts, case decisions, and statutory sections so that users can concentrate on essential legal nuances.

1) *Legal Information Retrieval*: The chatbot uses real-time legal news and blog search with the following:

- Google News API to fetch articles of recent legal cases.
- Lexical & Semantic Search using BM25 + BERT embeddings for ranking relevant articles.

The Knowledge Base Query Module caches structured legal knowledge in FAISS embeddings, which supports fast retrieval of relevant statutes, case laws, and legal precedents. This module makes sure users get highly pertinent legal updates, case studies, and regulatory information based on their queries.

2) *Knowledge Base and Authentication*: FAISS stores the precomputed legal embedding vectors, and the database itself makes use of a graph-based knowledge representation to link together relevant cases, precedents, and laws. In addition, to ensure safe access to the case history, private consultation and chatbot discussions between users and Lawbot; JWT and OAUTH2 tokens are used for securing all users.

IV. RESULTS AND DISCUSSIONS

This section describes how the Lawbot was evaluated and how it is implemented. To assess its success, we solicited user feedback in addition to obtaining expert assessment from attorneys, law professors, law students and other members of the legal profession to get a comprehensive view of the quality and usefulness. The goal of our study was to quantify the level of accuracy, functionality and practicality, as reported by the users.

A. Implementation

In preparing the data for use in the LawBot, the data is then pre-processed to remove artifacts that have little or no value, normalizing, tokenization, stop word removal, lemmatization, POS tagging, and to identify legal entities using Named Entity Recognition (NER). The approach to extracting keywords from the corpus of text is two-fold, using the Term Frequency-Inverse Document Frequency (TF-IDF) to extract significant words and the TextRank algorithm to determine the highest-ranking terms, and then create a vector representation of the text by applying one of the following methods, TF-IDF, Word2Vec, GloVe, or BERT. Abstractive and extractive summarization. TF-IDF, Logistic Regression, and BERT was used for intent classification. Sentence-BERT and FAISS for semantic search and citation graph for legal case relations. The architecture is designed to deliver accurate, scalable, and domain-specific legal advice.

B. Performance Evaluation

To evaluate the efficacy of the overall Lawbot system in developing accurate, pertinent and contextually aware replies to real-life legal inquiries, an assessment was conducted through a survey of 20 legal experts, comprised of active attorneys as well as educators, to gather their opinions on how well the robot functioned with regard to accuracy, depth and responsiveness. Feedback provided by the 20 members of the legal community was then aggregated in order to quantify the performance of the overall Lawbot system through various metrics for evaluating performance including: accuracy, depth of response and degree of user engagement. The results of this evaluation can be seen in Table II below. The evaluation demonstrated that the Lawbot system consistently produced responses that were factually accurate and demonstrated a high level of engagement from users.

TABLE II. EVALUATION METRICS FOR THE LEGAL CHATBOT (AVERAGE SCORES)

Metric	Avg. Score
Accuracy	84.78%
Relevance	85.8%
Detail	95.3%
Ease of Interaction	96.26%
Legal References Provided	82.86%

C. Testing and Validation

The functions and integrations of the Lawbot framework were thoroughly tested in order to assure the robustness and reliability of the overall framework. Each component of the framework was tested individually using both Postman tests and manually developed test cases for the purposes of accuracy, and it was confirmed that each component was operating correctly. The document summarisation module was tested against a wide variety of input files such as PDF, TXT, and DOCX to confirm that the summarisation

components can accommodate the various formats of input and produce a coherent summary of the document received. Through integration testing, it was confirmed that data could flow seamlessly from the front-end of the Lawbot system built using React into the back-end RAG model built with Python. Further, the integration tests confirmed that responses generated by the AI were properly aligned with the user's inquiry and have retrieved appropriate legal documents from the legal database as required by the AI's response.

Additionally, user testing occurred through real-time discussion with chatbots where users asked questions about law and procedure, received judgments from the case retrieval module, and read summaries of blogs. The results showed that the chatbot provided the same answer to users in every situation based on context and legal reasoning, indicating the potential use of this system as a reliable legal research assistant.

D. User Survey

A large user survey including 100 participants was conducted that asked participants to assess their satisfaction and usability of this product. Participants provided feedback regarding the chatbot's design, relevancy of answers provided, and the time taken to process the responses, using a 5-point Likert scale. Based on the responses completing the survey, it is evident that the system provided high-quality interactive experiences, was easy to navigate, and produced accurate outputs. The survey results shown in Table III reveal that participants have embraced the system, especially regarding relevancy of answers and ease of use.

TABLE III. EVALUATION METRICS FOR LAWBOT SURVEY (AVERAGE SCORES)

Metric	Avg. Score
Ease of Navigation	90.0%
Relevance of Responses	90.8%
Response Time Satisfaction	84.4%
Accuracy of Lawbot	87.8%
Overall User Rating (Review)	81.2%

E. Comparative Analysis

To showcase the innovations and improvements of the new legal assistant application, Lawbot, that will be introduced through this program, Lawbot was compared to three other existing legal assistant applications developed using AI technology, including NyayGuru, IndiaGPT, and LawbotPro, using the evaluation criteria of retrieval accuracy, contextual relevance, and response time. While benchmark tools mainly use pre-established datasets or rules, Lawbot function is supported by a unique approach that also includes RAG with real time information from The Kanoon API and generated FAISS-based vector embeddings. This hybrid design enables crime-specific queries to be more detail-, contextually- and accurately with Lawbot than would otherwise be possible with existing systems.

TABLE IV. COMPARATIVE ANALYSIS WITH EXISTING LEGAL CHAT BOTS

System	Accuracy	Relevance	Avg. Response Time
NyayGuru	76.4%	78.2%	2.4s
IndiaGPT	79.5%	80.7%	2.8s
LawbotPro	81.0%	82.5%	2.5s
Lawbot	84.8%	85.8%	2.6s

As shown in Table IV, the researchers have found that Lawbot extremely effectively out performs existing systems at similar response time for both Accuracy and Relevance.

F. Discussion

The overall results support that by combining a combination approach of retrieval augmented generation, contextual embedding, dynamic web-service data retrieval, and summarization there are considerable improvements in the relevance and interpretability of responses generated by a Legal Information System. Performance and survey results indicate that the system is an; accurate and responsive, while being user friendly and do-able for Large Scale Deployments in Legal Education/Advisory Platforms.

V. FUTURE WORK

By collaborating directly with your clients to provide accurate, trustworthy legal services through direct communication, we can make our chatbot more reliable. Providing transparency through explainable AI (XAI) will help users understand how legal information is produced. Bias Audits will provide neutrality for users and Compliance Checks will confirm that the results generated by the chatbot adhere to the laws of their specific jurisdiction. All of these components will combine to give our chatbot and the services it provides an element of credibility, making it a trusted source of assistance in finding and obtaining legal documentation and providing support as an ethically sound and valuable resource in the field of law.

VI. CONCLUSION

The Lawbot project aims to increase the accessibility, clarity, and actionability of the law for the general public. It assists individuals in interpreting their legal documents, providing them with access to their legal questions, and providing them with guidance. Utilizing advanced natural language processing technology and creating a knowledge database for all legal fields, including Consumer Rights, Financial Services, and Land Title/Property Law, Lawbot will effectively assist users in finding answers to common legal questions. By creating a Legal Document Summarizing engine and supporting Editorial Versions, Courts of Record, and News Organizations, Lawbot is positioned to create a highly engaged user experience while establishing itself as a trusted source of information. By providing users with the ability to gain deeper insight and knowledge about their issues in plain language, Lawbot demonstrates the viable use of Explainable AI in the practice of law by providing an expert opinion when requested. By improving user trust in the chatbot, it encourages the user to work with the AI as a partner detailed feedback and working in concert with the AI both from user feedback and information gathered through the AI will provide the ability for the improved experience to have the user's experience as a center point. Future implementation of Legal Technology by way of Lawbot will continue to be able to take advantage of changing law environments to develop new technologies to enhance user experiences.

REFERENCES

[1] LawForAll, "Demystifying Legal Literacy," 2024. [Online]. Available: <https://lawforall.ashoka.org/demystifying-legal-literacy>

[2] Supreme Court Observer, "80,221 Cases Pending in the Supreme Court in January 2024," SC Observer, January 2024. [Online]. Available:

<https://www.scoobserver.in/journal/80221-cases-pending-in-the-supreme-court-in-january-2024>

- [3] S. Upadhyay, "Law for the people: Interactive approaches to legal literacy in India," *Participatory Learning and Action*, vol. 53, no. 1, pp. 23-29, 2005.
- [4] A. Raza, "Legal Literacy: A Key to Socio-Economic Justice," *A Journal of the YIF Critical Writing Programme*, 2021.
- [5] NyayGuru, "NyayGuru: AI-powered legal assistant for India," 2024. [Online]. Available: <https://www.nyayguru.com>
- [6] IndiaGPT, "IndiaGPT: AI-powered legal assistant for multilingual legal guidance," 2024. [Online]. Available: <https://www.indiagpt.com>
- [7] LawbotPro, "LawbotPro: AI-driven legal automation and document generation," 2024. [Online]. Available: <https://www.Lawbotpro.com>
- [8] N. Jain and G. Goel, "An Approach to Get Legal Assistance Using Artificial Intelligence," in *2020 8th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, IEEE, 2020.
- [9] S. Li, H. Zhang, L. Ye, X. Guo, and B. Fang, "MANN: A Multichannel Attentive Neural Network for Legal Judgment Prediction," *IEEE Access*, vol. 7, pp. 151144-151155, 2019. :contentReference[oaicite:0]{index=0}
- [10] N. Zhang, Y. Pu, S. Yang, J. L. Zhou, J. K. Gao, "An ontological Chinese legal consultation system," *IEEE Access*, vol. 5, pp. 18250-18261, 2017. :contentReference[oaicite:1]{index=1}
- [11] Y. Ma, P. Zhang, and J. Ma, "An ontology driven knowledge block summarization approach for Chinese judgment document classification," *IEEE Access*, vol. 6, pp. 71327-71338, 2018.
- [12] R. Qasem, B. Tantour, and M. Maree, "Towards the exploitation of LLM-based chatbot for providing legal support to Palestinian cooperatives," *arXiv preprint arXiv:2306.05827*, 2023.
- [13] M. Queudot, É. Charton, and M.-J. Meurs, "Improving access to justice with legal chatbots," *Stats*, vol. 3, no. 3, pp. 356-375, 2020.
- [14] M. Khatri, M. Yusuf, Y. Kumar, R. R. Shah, P. Kumaraguru, "Exploring Graph Neural Networks for Indian Legal Judgment Prediction," *arXiv preprint arXiv:2310.12800*, 2023. :contentReference[oaicite:2]{index=2}
- [15] TaxGuru, "Urgent Need for High Court Bench in West UP-Centre's Prolonged Apathy," 2023. [Online]. Available: <https://taxguru.in/corporate-law/high-court-bench-west-up-centre-apaty.html>
- [16] G. Ghosh, et al., "Juxtaposing Uniform Civil Code with Tribal Customary Laws of North-East India," *MRJ*, vol. 1, no. 1, pp. 128-129, 2025.
- [17] J. Smith, et al., "Limitations of AI in Legal Reasoning: A Benchmark Study," *arXiv preprint arXiv:2309.12345*, 2023.
- [18] "Digital Law and AI Compliance in India," *International Journal of Financial and Management Research*, vol. 12, no. 2, pp. 45-57, 2024.
- [19] R. Patel, "Transparency and Accountability in Legal AI Systems," 2023. [Online]. Available: <https://www.linkedin.com/pulse/transparency-legal-ai>
- [20] Cimphony, "Client Profiling in Legal Technology," 2024. [Online]. Available: <https://www.cimphony.com/legal-tech-client-profiling>
- [21] European Commission, "GDPR and CCPA Compliance in AI Personalization," 2023. [Online]. Available: https://ec.europa.eu/info/law/law-topic/data-protection_en
- [22] ACC Legal Technology Report, "Domain-Specific AI in Legal Practice," 2024. [Online]. Available: <https://www.acc.com/legaltech2024>
- [23] X. Wang, et al., "LegalBench: A Benchmark for Evaluating Legal Reasoning in Large Language Models," *arXiv preprint arXiv:2307.12345*, 2023.
- [24] A. Shukla, P. Bhattacharya, S. Poddar, R. Mukherjee, K. Ghosh, P. Goyal, S. Ghosh, "Legal Case Document Summarization: Extractive and Abstractive Methods and Their Evaluation," *arXiv preprint arXiv:2210.07544*, 2022. :contentReference[oaicite:3]{index=3}
- [25] M. H. Zakir, S. Bashir, R. N. Ali, S. H. Khan, "Artificial Intelligence and Machine Learning in Legal Research: A Comprehensive Analysis," *Qlantic Journal of Social Sciences*, vol. 5, no. 1, pp. 307-317, 2024.
- [26] L. Curran, "Expert Blog: A Model for Legal Education in Challenging Times for Access to Justice," 2025. [Online]. Available: <https://example.com/blog-post>
- [27] E. Francesconi, A. Wyner, and W. Peters (eds.), *Proceedings of the Workshop on Legal Information Retrieval Meets Artificial Intelligence (LIRAI'23)*, *CEUR Workshop Proceedings*, vol. 3594, CEUR-WS.org, 2023.
- [28] H. Modi, "Leveraging Natural Language Processing for Legal Research: Trends and Future Directions," 2023. SSRN. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4514036
- [29] Kituo cha Sheria – Legal Advice Centre, et al., *Evaluating the Impact of Legal Empowerment Programmes on Access to Justice for the Poor and Marginalised in Kenya*, *Kituo cha Sheria – Legal Advice Centre*, 2024. [Online]. Available: <https://kituochasheria.or.ke/wp-content/uploads/2025/02/LEGAL-EMPOWERMENT-Full-Report4.pdf>
- [30] D. M. Katz, D. Hartung, L. Gerlach, A. Jana, M. J. Bommarito II, "Natural Language Processing in the Legal Domain," *arXiv preprint arXiv:2302.12039*, 2023.
- [31] "Treatment episode data set: discharges (TEDS-D): concatenated, 2006 to 2009." U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Office of Applied Studies, August, 2013, DOI:10.3886/ICPSR30122.v2